

ML Data Preprocessing Pratik Ghaytidak (25143078)

Import libraries

```
In [3]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Import Dataset

```
In [14]: df = pd.read_csv('Employee.csv');
df.shape
df
```

```
Out[14]:
```

	Company	Age	Salary	Place	Country	Gender
0	TCS	20.0	NaN	Chennai	India	0
1	Infosys	30.0	NaN	Mumbai	India	0
2	TCS	35.0	2300.0	Calcutta	India	0
3	Infosys	40.0	3000.0	Delhi	India	0
4	TCS	23.0	4000.0	Mumbai	India	0
...	...	...	...	...	...	...
143	TCS	33.0	9024.0	Calcutta	India	1
144	Infosys	22.0	8787.0	Calcutta	India	1
145	Infosys	44.0	4034.0	Delhi	India	1
146	TCS	33.0	5034.0	Mumbai	India	1
147	Infosys	22.0	8202.0	Cochin	India	0

148 rows × 6 columns

Data Exploration

```
In [12]: df.duplicated().sum();
df.drop_duplicates(inplace = True);
df.shape
```

```
Out[12]: (144, 6)
```

```
In [15]: df.reset_index(inplace = True)
df.drop('index',axis = 1, inplace = True)
```

df

Out[15]:

	Company	Age	Salary	Place	Country	Gender
0	TCS	20.0	NaN	Chennai	India	0
1	Infosys	30.0	NaN	Mumbai	India	0
2	TCS	35.0	2300.0	Calcutta	India	0
3	Infosys	40.0	3000.0	Delhi	India	0
4	TCS	23.0	4000.0	Mumbai	India	0
...	...	...	...	...	...	...
143	TCS	33.0	9024.0	Calcutta	India	1
144	Infosys	22.0	8787.0	Calcutta	India	1
145	Infosys	44.0	4034.0	Delhi	India	1
146	TCS	33.0	5034.0	Mumbai	India	1
147	Infosys	22.0	8202.0	Cochin	India	0

148 rows × 6 columns

In [17]: df.head(5)

Out[17]:

	Company	Age	Salary	Place	Country	Gender
0	TCS	20.0	NaN	Chennai	India	0
1	Infosys	30.0	NaN	Mumbai	India	0
2	TCS	35.0	2300.0	Calcutta	India	0
3	Infosys	40.0	3000.0	Delhi	India	0
4	TCS	23.0	4000.0	Mumbai	India	0

In [18]: df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 148 entries, 0 to 147
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Company    140 non-null    object
1   Age        130 non-null    float64
2   Salary     124 non-null    float64
3   Place      134 non-null    object
4   Country    148 non-null    object
5   Gender     148 non-null    int64
dtypes: float64(2), int64(1), object(3)
memory usage: 7.1+ KB
```

```
In [19]: df.isna().sum()
```

```
Out[19]: Company      8
         Age         18
         Salary      24
         Place       14
         Country      0
         Gender       0
         dtype: int64
```

```
In [21]: df.columns
```

```
Out[21]: Index(['Company', 'Age', 'Salary', 'Place', 'Country', 'Gender'], dtype='object')
```

```
In [23]: unique = df.nunique()
         unique
```

```
Out[23]: Company      6
         Age         29
         Salary      40
         Place       11
         Country      1
         Gender       2
         dtype: int64
```

```
In [24]: df['Company'].unique()
```

```
Out[24]: array(['TCS', 'Infosys', 'CTS', nan, 'Tata Consultancy Services',
                'Cognizant', 'Infosys Pvt Lmt'], dtype=object)
```

```
In [28]: df['Company'] = df['Company'].replace('Tata Consultancy Services', 'TCS')
         df['Company'] = df['Company'].replace('Infosys Pvt Lmt', 'Infosys')
         df['Company'] = df['Company'].replace('CTS', 'Cognizant')
         df['Company'] = df['Company'].replace('Cognizant', 'Cognizant')
         df['Company'].unique()
```

```
Out[28]: array(['TCS', 'Infosys', 'Cognizant', nan], dtype=object)
```

```
In [31]: df['Age'].unique()
```

```
Out[31]: array([20., 30., 35., 40., 23., nan, 34., 45., 18., 22., 32., 37., 50.,
                21., 46., 36., 26., 41., 24., 25., 43., 19., 38., 51., 31., 44.,
                33., 17., 0., 54.])
```

```
In [32]: df['Salary'].unique()
```

```
Out[32]: array([ nan, 2300., 3000., 4000., 5000., 6000., 7000., 8000., 9000.,
                1089., 1234., 3030., 3045., 3184., 4824., 5835., 7084., 8943.,
                8345., 9284., 9876., 2034., 7654., 2934., 4034., 5034., 8202.,
                9024., 4345., 6544., 6543., 3234., 4324., 5435., 5555., 8787.,
                3454., 5654., 5009., 5098., 3033.])
```

```
In [33]: df['Place'].unique()
```

```
Out[33]: array(['Chennai', 'Mumbai', 'Calcutta', 'Delhi', 'Podicherry', 'Cochin',
              nan, 'Noida', 'Hyderabad', 'Bhopal', 'Nagpur', 'Pune'],
              dtype=object)
```

```
In [34]: df['Place'] = df['Place'].replace('Podicherry', 'Pondicherry')
```

```
In [35]: df['Place'].unique()
```

```
Out[35]: array(['Chennai', 'Mumbai', 'Calcutta', 'Delhi', 'Pondicherry', 'Cochin',
              nan, 'Noida', 'Hyderabad', 'Bhopal', 'Nagpur', 'Pune'],
              dtype=object)
```

```
In [36]: df['Country'].unique
```

```
Out[36]: <bound method Series.unique of 0      India
1      India
2      India
3      India
4      India
...
143    India
144    India
145    India
146    India
147    India
Name: Country, Length: 148, dtype: object>
```

```
In [37]: df['Gender'].unique
```

```
Out[37]: <bound method Series.unique of 0      0
1      0
2      0
3      0
4      0
..
143    1
144    1
145    1
146    1
147    0
Name: Gender, Length: 148, dtype: int64>
```

Statistical analysis

```
In [38]: df.describe()
```

Out[38]:

	Age	Salary	Gender
count	130.000000	124.000000	148.000000
mean	30.484615	5312.467742	0.222973
std	11.096640	2573.764683	0.417654
min	0.000000	1089.000000	0.000000
25%	22.000000	3030.000000	0.000000
50%	32.500000	5000.000000	0.000000
75%	37.750000	8000.000000	0.000000
max	54.000000	9876.000000	1.000000

In [39]: df.head(5)

Out[39]:

	Company	Age	Salary	Place	Country	Gender
0	TCS	20.0	NaN	Chennai	India	0
1	Infosys	30.0	NaN	Mumbai	India	0
2	TCS	35.0	2300.0	Calcutta	India	0
3	Infosys	40.0	3000.0	Delhi	India	0
4	TCS	23.0	4000.0	Mumbai	India	0

Data Cleaning

In [44]: df1 = pd.read_csv('Employee.csv')
df1.describe()

Out[44]:

	Age	Salary	Gender
count	130.000000	124.000000	148.000000
mean	30.484615	5312.467742	0.222973
std	11.096640	2573.764683	0.417654
min	0.000000	1089.000000	0.000000
25%	22.000000	3030.000000	0.000000
50%	32.500000	5000.000000	0.000000
75%	37.750000	8000.000000	0.000000
max	54.000000	9876.000000	1.000000

Age Outlier Detection

```
In [45]: q1 = df['Age'].quantile(0.25)
q1
```

```
Out[45]: np.float64(22.0)
```

```
In [46]: q3 = df['Age'].quantile(0.75)
q3
```

```
Out[46]: np.float64(37.75)
```

```
In [47]: iqr = q3 - q1
iqr
```

```
Out[47]: np.float64(15.75)
```

```
In [48]: lower = q1 - 1.5 * iqr
lower
```

```
Out[48]: np.float64(-1.625)
```

```
In [49]: upper = q3 + 1.5 * iqr
upper
```

```
Out[49]: np.float64(61.375)
```

```
In [68]: df[(df['Age'] < lower) | (df['Age'] < upper)]
```

```
Out[68]:
```

	Company	Age	Salary	Place	Country	Gender
0	TCS	20.0	NaN	Chennai	India	0
1	Infosys	30.0	NaN	Mumbai	India	0
2	TCS	35.0	2300.0	Calcutta	India	0
3	Infosys	40.0	3000.0	Delhi	India	0
4	TCS	23.0	4000.0	Mumbai	India	0
...	...	...	...	...	...	...
143	TCS	33.0	9024.0	Calcutta	India	1
144	Infosys	22.0	8787.0	Calcutta	India	1
145	Infosys	44.0	4034.0	Delhi	India	1
146	TCS	33.0	5034.0	Mumbai	India	1
147	Infosys	22.0	8202.0	Cochin	India	0

130 rows × 6 columns

Salary Outlier Detection

```
In [59]: q1 = df['Salary'].quantile(0.25)
          q1
```

```
Out[59]: np.float64(3030.0)
```

```
In [60]: q3 = df['Salary'].quantile(0.75)
          q3
```

```
Out[60]: np.float64(8000.0)
```

```
In [61]: iqr = q3 - q1
          iqr
```

```
Out[61]: np.float64(4970.0)
```

```
In [62]: lower = q1 - 1.5 * iqr
lower
```

```
Out[62]: np.float64(-4425.0)
```

```
In [63]: upper = q3 + 1.5 * iqr
         upper
```

```
Out[63]: np.float64(15455.0)
```

```
In [69]: df[(df['Salary'] < lower) | (df['Salary'] > upper)]
```

Out[69]: **Company** **Age** **Salary** **Place** **Country** **Gender**

```
In [70]: df.isna().sum()
```

```
Out[70]: Company      8
         Age          18
         Salary       24
         Place        14
         Country       0
         Gender        0
         dtype: int64
```

```
In [73]: df['Company'].mode()[0]
```

```
Out[73]: 'TCS'
```

```
In [76]: df['Company'] = df['Company'].fillna(df['Company'].mode()[0])
```

```
In [77]: df.isna().sum()
```

```
Out[77]: Company    0
         Age       18
         Salary    24
         Place     14
         Country    0
         Gender     0
         dtype: int64
```

```
In [90]: df[df['Age']==0]
```

```
Out[90]:   Company  Age  Salary  Place  Country  Gender
```

```
In [92]: df['Age'] = df['Age'].replace(0,np.nan)
```

```
In [93]: df.isna().sum()
```

```
Out[93]: Company    0
         Age       24
         Salary    24
         Place     14
         Country    0
         Gender     0
         dtype: int64
```

```
In [94]: df.describe()
```

```
Out[94]:
```

	Age	Salary	Gender
count	124.000000	124.000000	148.000000
mean	31.959677	5312.467742	0.222973
std	9.034171	2573.764683	0.417654
min	17.000000	1089.000000	0.000000
25%	23.000000	3030.000000	0.000000
50%	33.000000	5000.000000	0.000000
75%	40.000000	8000.000000	0.000000
max	54.000000	9876.000000	1.000000

```
In [97]: rounded_mean = round(df['Age'].mean(),0)
         rounded_mean
```

```
Out[97]: np.float64(32.0)
```

```
In [98]: df['Age'] = df['Age'].fillna(rounded_mean)
```

```
In [99]: df.isna().sum()
```



```
Out[99]: Company    0
         Age        0
         Salary    24
         Place     14
         Country    0
         Gender     0
         dtype: int64
```

```
In [101... rounded_salary = round(df['Salary'].mean(),0)
rounded_salary
```

```
Out[101... np.float64(5312.0)
```

```
In [102... df['Salary'] = df['Salary'].fillna(rounded_salary)
```

```
In [103... df.isna().sum()
```

```
Out[103... Company    0
         Age        0
         Salary    0
         Place     14
         Country    0
         Gender     0
         dtype: int64
```

```
In [104... df['Place'].mode()
```

```
Out[104... 0    Mumbai
         Name: Place, dtype: object
```

```
In [106... df['Place'] = df['Place'].fillna(df['Place'].mode()[0])
```

```
In [107... df.isna().sum()
```

```
Out[107... Company    0
         Age        0
         Salary    0
         Place     0
         Country    0
         Gender     0
         dtype: int64
```

```
In [108... df
```

Out[108...

	Company	Age	Salary	Place	Country	Gender
0	TCS	20.0	5312.0	Chennai	India	0
1	Infosys	30.0	5312.0	Mumbai	India	0
2	TCS	35.0	2300.0	Calcutta	India	0
3	Infosys	40.0	3000.0	Delhi	India	0
4	TCS	23.0	4000.0	Mumbai	India	0
...	...	...	...	...	...	...
143	TCS	33.0	9024.0	Calcutta	India	1
144	Infosys	22.0	8787.0	Calcutta	India	1
145	Infosys	44.0	4034.0	Delhi	India	1
146	TCS	33.0	5034.0	Mumbai	India	1
147	Infosys	22.0	8202.0	Cochin	India	0

148 rows × 6 columns

Data Analysis

In [111...

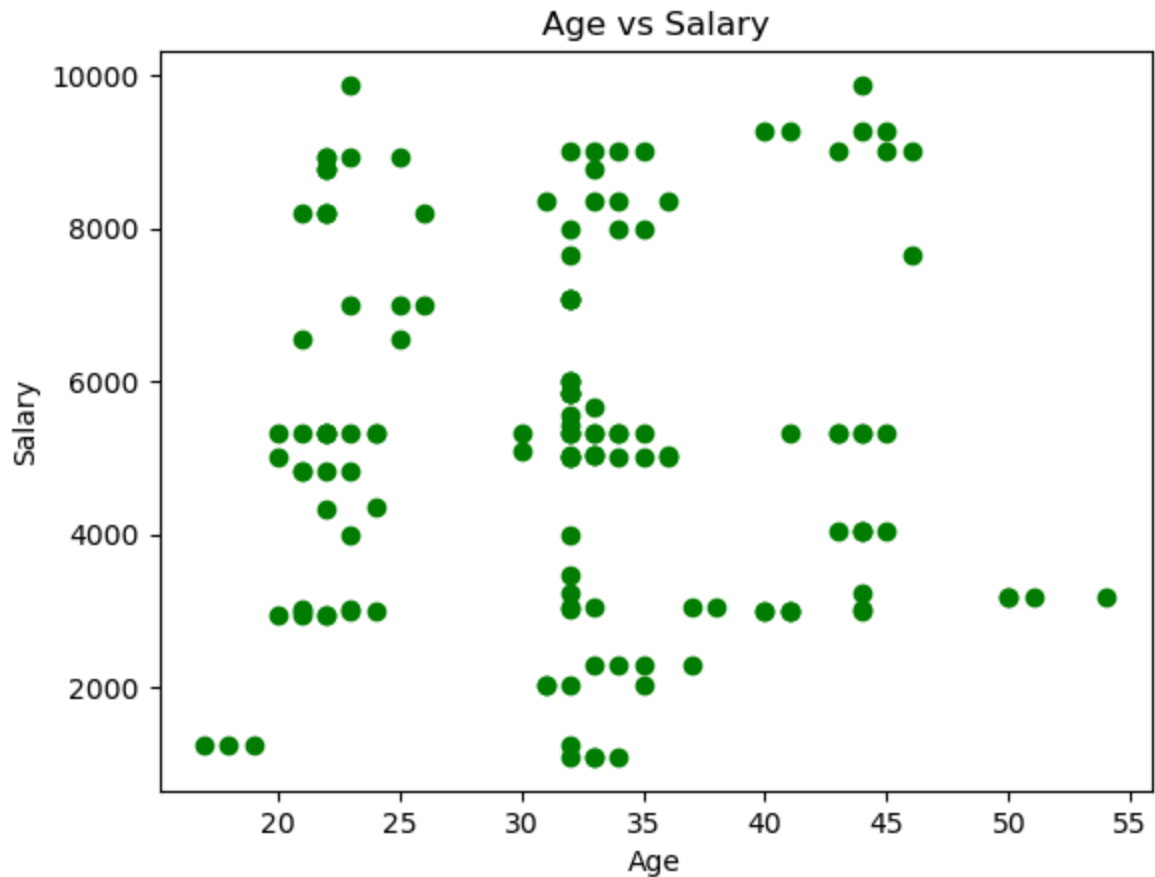
```
df[(df['Age'] > 40) & (df['Salary'] < 5000)]
```

Out[111...

	Company	Age	Salary	Place	Country	Gender
21	Infosys	50.0	3184.0	Delhi	India	0
32	Infosys	45.0	4034.0	Calcutta	India	0
39	Infosys	41.0	3000.0	Mumbai	India	0
50	Infosys	41.0	3000.0	Chennai	India	0
57	Infosys	51.0	3184.0	Hyderabad	India	0
68	Infosys	43.0	4034.0	Mumbai	India	0
75	Infosys	44.0	3000.0	Cochin	India	0
86	Infosys	41.0	3000.0	Delhi	India	0
93	Infosys	54.0	3184.0	Mumbai	India	0
104	Infosys	44.0	4034.0	Delhi	India	0
122	Infosys	44.0	3234.0	Mumbai	India	0
129	Infosys	50.0	3184.0	Calcutta	India	0
138	Cognizant	44.0	3033.0	Cochin	India	0
140	Infosys	44.0	4034.0	Hyderabad	India	0
145	Infosys	44.0	4034.0	Delhi	India	1

In [114...

```
plt.scatter(df['Age'],df['Salary'],color = 'g')
plt.xlabel("Age")
plt.ylabel("Salary")
plt.title("Age vs Salary")
plt.show()
```



Observations from scatter plot

```
In [115...] df['Place'].value_counts()
```

```
Out[115...] Place
Mumbai          51
Calcutta        33
Chennai         14
Delhi           14
Cochin          13
Noida           8
Hyderabad       8
Pondicherry     3
Pune            2
Bhopal           1
Nagpur           1
Name: count, dtype: int64
```

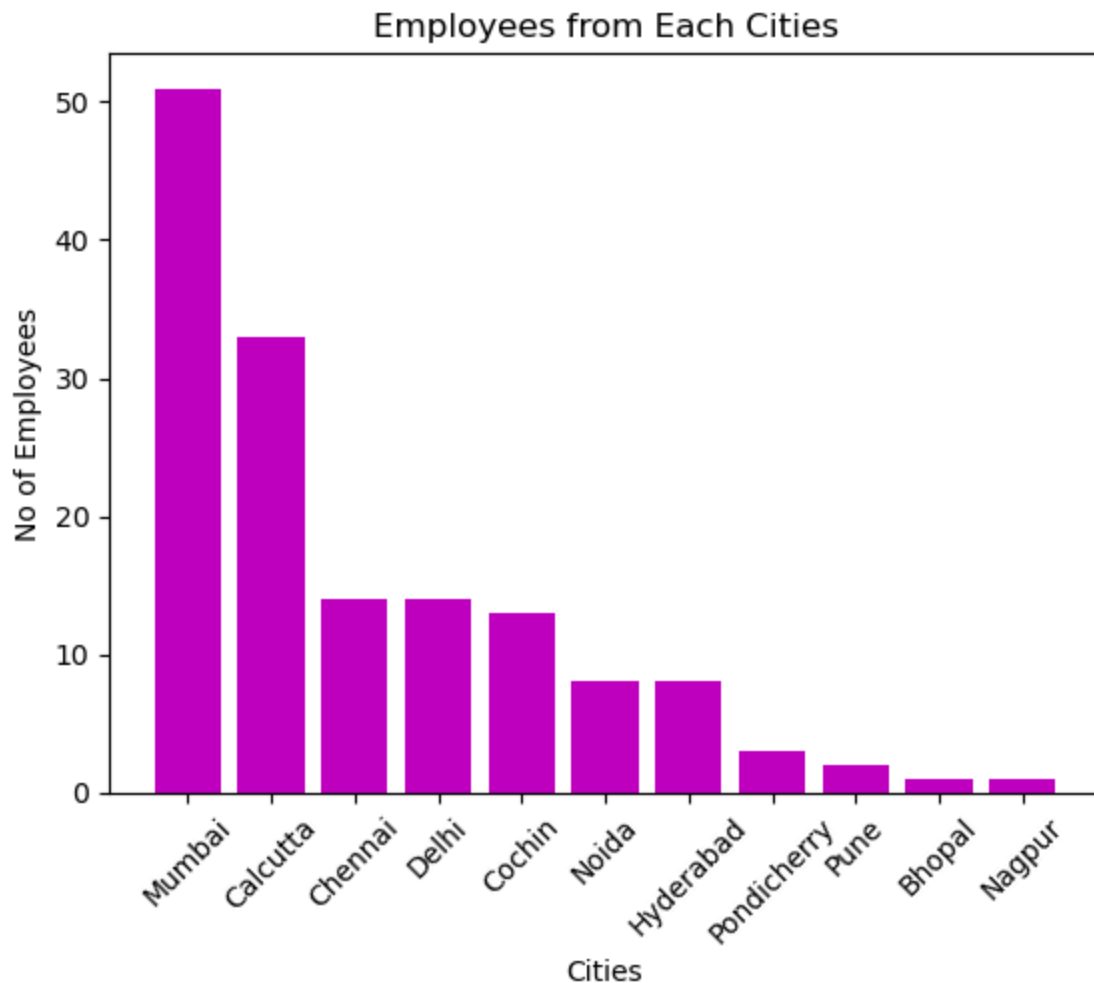
```
In [116...] x = df['Place'].value_counts().index
x
```

```
Out[116...] Index(['Mumbai', 'Calcutta', 'Chennai', 'Delhi', 'Cochin', 'Noida',
                  'Hyderabad', 'Pondicherry', 'Pune', 'Bhopal', 'Nagpur'],
                  dtype='object', name='Place')
```

```
In [117...] y = df['Place'].value_counts().values
y
```

Out[117...] array([51, 33, 14, 14, 13, 8, 8, 3, 2, 1, 1])

```
In [119...] plt.bar(x,y,color = 'm')
plt.xlabel("Cities")
plt.ylabel("No of Employees")
plt.title("Employees from Each Cities")
plt.xticks(rotation = 45)
plt.show()
```



Data Encoding

```
In [120...] df.head(1)
```

```
Out[120...]
   Company  Age  Salary  Place  Country  Gender
0      TCS  20.0  5312.0  Chennai    India      0
```

```
In [121...] df['Gender'] = df['Gender'].replace(0,'M')
```

```
In [122...] df['Gender'] = df['Gender'].replace(1,'F')
```

```
In [123...] df.head()
```

Out[123...

	Company	Age	Salary	Place	Country	Gender
0	TCS	20.0	5312.0	Chennai	India	M
1	Infosys	30.0	5312.0	Mumbai	India	M
2	TCS	35.0	2300.0	Calcutta	India	M
3	Infosys	40.0	3000.0	Delhi	India	M
4	TCS	23.0	4000.0	Mumbai	India	M

In [124...

```
df['Gender'].value_counts()
```

Out[124...

```
Gender
M    115
F     33
Name: count, dtype: int64
```

In [125...

```
df.describe()
```

Out[125...

	Age	Salary
count	148.000000	148.000000
mean	31.966216	5312.391892
std	8.263859	2354.305009
min	17.000000	1089.000000
25%	23.000000	3149.250000
50%	32.000000	5312.000000
75%	36.000000	7084.000000
max	54.000000	9876.000000

In [126...

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 148 entries, 0 to 147
Data columns (total 6 columns):
#   Column    Non-Null Count  Dtype
---  -
0   Company   148 non-null    object
1   Age       148 non-null    float64
2   Salary    148 non-null    float64
3   Place     148 non-null    object
4   Country   148 non-null    object
5   Gender    148 non-null    object
dtypes: float64(2), object(4)
memory usage: 7.1+ KB
```

```
In [127... df.drop('Country',axis = 1, inplace = True)
df
```

```
Out[127...
   Company  Age  Salary  Place  Gender
0      TCS  20.0   5312.0  Chennai      M
1  Infosys  30.0   5312.0  Mumbai      M
2      TCS  35.0   2300.0  Calcutta      M
3  Infosys  40.0   3000.0    Delhi      M
4      TCS  23.0   4000.0  Mumbai      M
...      ...   ...      ...      ...
143     TCS  33.0   9024.0  Calcutta      F
144  Infosys  22.0   8787.0  Calcutta      F
145  Infosys  44.0   4034.0    Delhi      F
146     TCS  33.0   5034.0  Mumbai      F
147  Infosys  22.0   8202.0   Cochin      M
```

148 rows × 5 columns

OneHotEncoding

```
In [128... df['Company'].unique()
```

```
Out[128... array(['TCS', 'Infosys', 'Cognizant'], dtype=object)
```

```
In [129... df['Place'].unique()
```

```
Out[129... array(['Chennai', 'Mumbai', 'Calcutta', 'Delhi', 'Pondicherry', 'Cochin',
      'Noida', 'Hyderabad', 'Bhopal', 'Nagpur', 'Pune'], dtype=object)
```

```
In [130... df['Gender'].unique()
```

```
Out[130... array(['M', 'F'], dtype=object)
```

```
In [131... from sklearn.preprocessing import OneHotEncoder
```

```
In [132... ohe=OneHotEncoder()
```

```
In [133... ohe
```

```
Out[133...
▼ OneHotEncoder ⓘ ?
  ► Parameters
```

```
In [134... df_array=ohe.fit_transform(df[['Company','Place','Gender']]).toarray()
df_array
```

```
Out[134... array([[0., 0., 1., ..., 0., 0., 1.],
        [0., 1., 0., ..., 0., 0., 1.],
        [0., 0., 1., ..., 0., 0., 1.],
        ...,
        [0., 1., 0., ..., 0., 1., 0.],
        [0., 0., 1., ..., 0., 1., 0.],
        [0., 1., 0., ..., 0., 0., 1.]], shape=(148, 16))
```

```
In [135... ohe.categories_
```

```
Out[135... [array(['Cognizant', 'Infosys', 'TCS'], dtype=object),
          array(['Bhopal', 'Calcutta', 'Chennai', 'Cochin', 'Delhi', 'Hyderabad',
                'Mumbai', 'Nagpur', 'Noida', 'Pondicherry', 'Pune'], dtype=object),
          array(['F', 'M'], dtype=object)]
```

```
In [136... categories=[np.array(['Cognizant','Infosys','TCS']),
                  np.array(['Bhopal','Calcutta','Chennai','Cochin','Delhi','Hyderabad','M
                  np.array(['F','M'])]

single_list=[innerlistone for main_array in categories for innerlistone in main_arr
single_list
```

```
Out[136... [np.str_('Cognizant'),
            np.str_('Infosys'),
            np.str_('TCS'),
            np.str_('Bhopal'),
            np.str_('Calcutta'),
            np.str_('Chennai'),
            np.str_('Cochin'),
            np.str_('Delhi'),
            np.str_('Hyderabad'),
            np.str_('Mumbai'),
            np.str_('Nagpur'),
            np.str_('Noida'),
            np.str_('Pondicherry'),
            np.str_('Pune'),
            np.str_('F'),
            np.str_('M')]
```

```
In [137... df_new=pd.DataFrame(df_array,columns=single_list)
df_new
```


Out[137...

	Cognizant	Infosys	TCS	Bhopal	Calcutta	Chennai	Cochin	Delhi	Hyderabad	Murr
0	0.0	0.0	1.0	0.0	0.0	1.0	0.0	0.0		0.0
1	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0		0.0
2	0.0	0.0	1.0	0.0	1.0	0.0	0.0	0.0		0.0
3	0.0	1.0	0.0	0.0	0.0	0.0	0.0	1.0		0.0
4	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0		0.0
...	...	...	...	...	...	...	...	...		...
143	0.0	0.0	1.0	0.0	1.0	0.0	0.0	0.0		0.0
144	0.0	1.0	0.0	0.0	1.0	0.0	0.0	0.0		0.0
145	0.0	1.0	0.0	0.0	0.0	0.0	0.0	1.0		0.0
146	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0		0.0
147	0.0	1.0	0.0	0.0	0.0	0.0	1.0	0.0		0.0

148 rows × 16 columns



In [138...

```
df_ml=pd.concat([df,df_new],axis=1)
df_ml
```

Out[138...

	Company	Age	Salary	Place	Gender	Cognizant	Infosys	TCS	Bhopal	Calcutta
0	TCS	20.0	5312.0	Chennai	M	0.0	0.0	1.0	0.0	0.0
1	Infosys	30.0	5312.0	Mumbai	M	0.0	1.0	0.0	0.0	0.0
2	TCS	35.0	2300.0	Calcutta	M	0.0	0.0	1.0	0.0	1.0
3	Infosys	40.0	3000.0	Delhi	M	0.0	1.0	0.0	0.0	0.0
4	TCS	23.0	4000.0	Mumbai	M	0.0	0.0	1.0	0.0	0.0
...	...	...	...	...	...	...	...	...	...	...
143	TCS	33.0	9024.0	Calcutta	F	0.0	0.0	1.0	0.0	1.0
144	Infosys	22.0	8787.0	Calcutta	F	0.0	1.0	0.0	0.0	1.0
145	Infosys	44.0	4034.0	Delhi	F	0.0	1.0	0.0	0.0	0.0
146	TCS	33.0	5034.0	Mumbai	F	0.0	0.0	1.0	0.0	0.0
147	Infosys	22.0	8202.0	Cochin	M	0.0	1.0	0.0	0.0	0.0

148 rows × 21 columns



LabelEncoder

```
In [34]: df.to_csv('Employee_cleaned.csv', index=False)
```

```
In [140]: from sklearn.preprocessing import LabelEncoder
```

```
In [141]: label=LabelEncoder()
print(label)
```

LabelEncoder()

```
In [142]: df_sample=pd.read_csv('Employee_cleaned.csv')
df_sample['Company']=label.fit_transform(df_sample['Company'])
```

```
In [143]: df_sample
```

```
Out[143]:
```

	Company	Age	Salary	Place	Gender
0	2	20.0	5312.0	Chennai	M
1	1	30.0	5312.0	Mumbai	M
2	2	35.0	2300.0	Calcutta	M
3	1	40.0	3000.0	Delhi	M
4	2	23.0	4000.0	Mumbai	M
...	...	...	...	...	...
143	2	33.0	9024.0	Calcutta	F
144	1	22.0	8787.0	Calcutta	F
145	1	44.0	4034.0	Delhi	F
146	2	33.0	5034.0	Mumbai	F
147	1	22.0	8202.0	Cochin	M

148 rows × 5 columns

StandardScaler

```
In [5]: df = pd.read_csv('Employee_cleaned.csv')
```

```
In [7]: data = df.iloc[:, [1,2]]
data
```

```
Out[7]:
```

	Age	Salary
0	20.0	5312.0
1	30.0	5312.0
2	35.0	2300.0
3	40.0	3000.0
4	23.0	4000.0
...	...	...
143	33.0	9024.0
144	22.0	8787.0
145	44.0	4034.0
146	33.0	5034.0
147	22.0	8202.0

148 rows × 2 columns

```
In [8]: from sklearn.preprocessing import StandardScaler
```

```
In [12]: scaler=StandardScaler().fit(data)
print(scaler)
```

StandardScaler()

```
In [13]: data_scaled = scaler.transform(data)
print(data_scaled)
```

[[-1.45293481e+00 -1.67022793e-04]
[-2.38737454e-01 -1.67022793e-04]
[3.68361226e-01 -1.28386965e+00]
[9.75459906e-01 -9.85532389e-01]
[-1.08867561e+00 -5.59336296e-01]
[4.10201811e-03 -1.33140204e-01]
[4.10201811e-03 2.93055889e-01]
[-1.08867561e+00 7.19251981e-01]
[2.46941490e-01 1.14544807e+00]
[1.58255859e+00 1.57164417e+00]
[-1.08867561e+00 -1.67022793e-04]
[2.46941490e-01 -1.79999312e+00]
[1.58255859e+00 -1.67022793e-04]
[-1.69577429e+00 -1.73819469e+00]
[9.75459906e-01 -9.85532389e-01]
[-1.08867561e+00 -9.85532389e-01]
[-1.08867561e+00 -9.72746506e-01]
[2.46941490e-01 -1.33140204e-01]
[-1.21009534e+00 -1.67022793e-04]
[4.10201811e-03 -1.67022793e-04]
[6.11200698e-01 -9.66353564e-01]
[2.18965727e+00 -9.07112308e-01]
[-1.33151508e+00 -2.08150716e-01]
[4.10201811e-03 2.22733534e-01]
[4.10201811e-03 7.55052453e-01]
[-1.08867561e+00 1.54735099e+00]
[2.46941490e-01 1.29248573e+00]
[1.58255859e+00 1.69268386e+00]
[-1.08867561e+00 1.94499194e+00]
[3.68361226e-01 -1.39723781e+00]
[1.70397832e+00 9.97984226e-01]
[-1.45293481e+00 -1.01366133e+00]
[1.58255859e+00 -5.44845629e-01]
[4.89780962e-01 -1.18649537e-01]
[-7.24416398e-01 1.23153968e+00]
[3.68361226e-01 1.58187287e+00]
[4.10201811e-03 -1.67022793e-04]
[3.68361226e-01 -1.67022793e-04]
[2.46941490e-01 -1.28386965e+00]
[1.09687964e+00 -9.85532389e-01]
[-9.67255870e-01 -4.12298644e-01]
[4.10201811e-03 -1.33140204e-01]
[4.10201811e-03 2.93055889e-01]
[-8.45836134e-01 7.19251981e-01]
[3.68361226e-01 1.14544807e+00]
[1.70397832e+00 1.57164417e+00]
[-9.67255870e-01 -1.67022793e-04]
[4.10201811e-03 -1.79999312e+00]
[1.33971911e+00 -1.67022793e-04]
[-1.57435455e+00 -1.73819469e+00]
[1.09687964e+00 -9.85532389e-01]
[-9.67255870e-01 -9.85532389e-01]
[-1.33151508e+00 -9.72746506e-01]
[3.68361226e-01 -1.33140204e-01]
[-1.33151508e+00 5.24906563e-01]
[4.10201811e-03 9.97984226e-01]

[7.32620434e-01 -9.66353564e-01]
[2.31107700e+00 -9.07112308e-01]
[-1.08867561e+00 -2.08150716e-01]
[4.10201811e-03 2.22733534e-01]
[4.10201811e-03 7.55052453e-01]
[-8.45836134e-01 1.54735099e+00]
[4.89780962e-01 1.29248573e+00]
[1.09687964e+00 1.69268386e+00]
[-8.45836134e-01 5.24480367e-01]
[-1.17317718e-01 -1.39723781e+00]
[1.09687964e+00 -1.67022793e-04]
[-1.33151508e+00 -1.01366133e+00]
[1.33971911e+00 -5.44845629e-01]
[4.10201811e-03 -1.18649537e-01]
[-1.33151508e+00 1.23153968e+00]
[2.46941490e-01 1.58187287e+00]
[-9.67255870e-01 -1.67022793e-04]
[2.46941490e-01 -1.67022793e-04]
[6.11200698e-01 -1.28386965e+00]
[1.46113885e+00 -9.85532389e-01]
[4.10201811e-03 -5.59336296e-01]
[4.10201811e-03 -1.33140204e-01]
[4.10201811e-03 2.93055889e-01]
[-7.24416398e-01 7.19251981e-01]
[4.10201811e-03 1.14544807e+00]
[1.33971911e+00 1.57164417e+00]
[-1.21009534e+00 -1.67022793e-04]
[1.25521754e-01 -1.79999312e+00]
[1.33971911e+00 -1.67022793e-04]
[-1.81719402e+00 -1.73819469e+00]
[1.09687964e+00 -9.85532389e-01]
[-1.33151508e+00 -9.85532389e-01]
[4.10201811e-03 -9.72746506e-01]
[4.89780962e-01 -1.33140204e-01]
[-1.33151508e+00 -1.67022793e-04]
[2.46941490e-01 -1.67022793e-04]
[4.10201811e-03 -9.66353564e-01]
[2.67533621e+00 -9.07112308e-01]
[-1.21009534e+00 -2.08150716e-01]
[4.10201811e-03 2.22733534e-01]
[4.10201811e-03 7.55052453e-01]
[-1.21009534e+00 1.54735099e+00]
[1.25521754e-01 1.29248573e+00]
[1.46113885e+00 1.69268386e+00]
[-1.21009534e+00 -1.67022793e-04]
[4.10201811e-03 -1.39723781e+00]
[1.46113885e+00 -1.67022793e-04]
[-1.21009534e+00 -1.01366133e+00]
[1.46113885e+00 -5.44845629e-01]
[1.25521754e-01 -1.18649537e-01]
[-1.21009534e+00 1.23153968e+00]
[4.10201811e-03 1.58187287e+00]
[1.25521754e-01 -1.67022793e-04]
[1.25521754e-01 -1.67022793e-04]
[1.25521754e-01 -1.28386965e+00]
[4.10201811e-03 -8.85802503e-01]

```

[-1.21009534e+00 -4.21248762e-01]
[ 4.10201811e-03  5.22550966e-02]
[ 4.10201811e-03  1.03398628e-01]
[-1.21009534e+00  1.48086440e+00]
[ 1.25521754e-01  1.48086440e+00]
[ 1.46113885e+00  1.94499194e+00]
[-1.21009534e+00 -1.67022793e-04]
[ 1.25521754e-01 -1.79999312e+00]
[ 1.46113885e+00 -1.67022793e-04]
[ 4.10201811e-03 -1.73819469e+00]
[ 1.46113885e+00 -8.85802503e-01]
[ 4.10201811e-03 -7.92039363e-01]
[-1.21009534e+00  1.48086440e+00]
[ 1.25521754e-01  1.45592041e-01]
[-1.45293481e+00 -1.29304439e-01]
[-2.38737454e-01 -9.13729866e-02]
[ 1.25521754e-01 -9.66353564e-01]
[ 2.18965727e+00 -9.07112308e-01]
[-1.33151508e+00 -2.08150716e-01]
[ 4.10201811e-03  2.22733534e-01]
[ 4.10201811e-03  7.55052453e-01]
[-1.21009534e+00  1.54735099e+00]
[-1.17317718e-01  1.29248573e+00]
[ 9.75459906e-01  1.69268386e+00]
[ 4.10201811e-03 -1.67022793e-04]
[-1.17317718e-01 -1.39723781e+00]
[ 1.46113885e+00 -9.71467918e-01]
[-1.21009534e+00 -1.01366133e+00]
[ 1.46113885e+00 -5.44845629e-01]
[ 1.25521754e-01 -1.18649537e-01]
[-1.21009534e+00  1.23153968e+00]
[ 1.25521754e-01  1.58187287e+00]
[-1.21009534e+00  1.48086440e+00]
[ 1.46113885e+00 -5.44845629e-01]
[ 1.25521754e-01 -1.18649537e-01]
[-1.21009534e+00  1.23153968e+00]]

```

```
In [14]: print(data_scaled.mean(axis=0))
```

```
[-1.66533454e-16  1.12522604e-16]
```

```
In [15]: print(data_scaled.std(axis=0))
```

```
[1.  1.]
```

```
In [16]: scaled_data_set = pd.DataFrame(data_scaled, columns=data.columns)
scaled_data_set
```

Out[16]:

	Age	Salary
0	-1.452935	-0.000167
1	-0.238737	-0.000167
2	0.368361	-1.283870
3	0.975460	-0.985532
4	-1.088676	-0.559336
...	...	...
143	0.125522	1.581873
144	-1.210095	1.480864
145	1.461139	-0.544846
146	0.125522	-0.118650
147	-1.210095	1.231540

148 rows × 2 columns

```
In [33]: scaled_data_set.to_csv("scaled_data_set.csv")
```

MinMaxScaler

```
In [17]: data = df.iloc[:,[1,2]]  
data
```

Out[17]:

	Age	Salary
0	20.0	5312.0
1	30.0	5312.0
2	35.0	2300.0
3	40.0	3000.0
4	23.0	4000.0
...	...	...
143	33.0	9024.0
144	22.0	8787.0
145	44.0	4034.0
146	33.0	5034.0
147	22.0	8202.0

148 rows × 2 columns

In [20]: `from sklearn.preprocessing import MinMaxScaler`In [22]: `scaler = MinMaxScaler().fit(data)`
`print(scaler)`

MinMaxScaler()

In [23]: `scaler.data_min_`Out[23]: `array([ 17., 1089.])`In [24]: `scaler.data_max_`Out[24]: `array([ 54., 9876.])`In [25]: `data.describe`


```
Out[25]: <bound method NDFrame.describe of      Age  Salary
0    20.0  5312.0
1    30.0  5312.0
2    35.0  2300.0
3    40.0  3000.0
4    23.0  4000.0
..     ...     ...
143   33.0  9024.0
144   22.0  8787.0
145   44.0  4034.0
146   33.0  5034.0
147   22.0  8202.0

[148 rows x 2 columns]>
```

```
In [26]: scaler.feature_range
```

```
Out[26]: (0, 1)
```

```
In [27]: data_scaled_mm = scaler.transform(data)
print(data_scaled_mm)
```

[0.08108108 0.48059634]
[0.35135135 0.48059634]
[0.48648649 0.13781723]
[0.62162162 0.21748037]
[0.16216216 0.33128485]
[0.40540541 0.44508934]
[0.40540541 0.55889382]
[0.16216216 0.6726983]
[0.45945946 0.78650279]
[0.75675676 0.90030727]
[0.16216216 0.48059634]
[0.45945946 0.]
[0.75675676 0.48059634]
[0.02702703 0.01650165]
[0.62162162 0.21748037]
[0.16216216 0.21748037]
[0.16216216 0.2208945]
[0.45945946 0.44508934]
[0.13513514 0.48059634]
[0.40540541 0.48059634]
[0.54054054 0.22260157]
[0.89189189 0.23842039]
[0.10810811 0.42505975]
[0.40540541 0.54011608]
[0.40540541 0.68225788]
[0.16216216 0.89382042]
[0.45945946 0.82576534]
[0.75675676 0.93262775]
[0.16216216 1.]
[0.48648649 0.10754524]
[0.78378378 0.74712644]
[0.08108108 0.20996927]
[0.75675676 0.33515421]
[0.51351351 0.44895869]
[0.24324324 0.80949129]
[0.48648649 0.90303858]
[0.40540541 0.48059634]
[0.48648649 0.48059634]
[0.45945946 0.13781723]
[0.64864865 0.21748037]
[0.18918919 0.3705474]
[0.40540541 0.44508934]
[0.40540541 0.55889382]
[0.21621622 0.6726983]
[0.48648649 0.78650279]
[0.78378378 0.90030727]
[0.18918919 0.48059634]
[0.40540541 0.]
[0.7027027 0.48059634]
[0.05405405 0.01650165]
[0.64864865 0.21748037]
[0.18918919 0.21748037]
[0.10810811 0.2208945]
[0.48648649 0.44508934]
[0.10810811 0.62080346]
[0.40540541 0.74712644]

[0.56756757 0.22260157]
[0.91891892 0.23842039]
[0.16216216 0.42505975]
[0.40540541 0.54011608]
[0.40540541 0.68225788]
[0.21621622 0.89382042]
[0.51351351 0.82576534]
[0.64864865 0.93262775]
[0.21621622 0.62068966]
[0.37837838 0.10754524]
[0.64864865 0.48059634]
[0.10810811 0.20996927]
[0.7027027 0.33515421]
[0.40540541 0.44895869]
[0.10810811 0.80949129]
[0.45945946 0.90303858]
[0.18918919 0.48059634]
[0.45945946 0.48059634]
[0.54054054 0.13781723]
[0.72972973 0.21748037]
[0.40540541 0.33128485]
[0.40540541 0.44508934]
[0.40540541 0.55889382]
[0.24324324 0.6726983]
[0.40540541 0.78650279]
[0.7027027 0.90030727]
[0.13513514 0.48059634]
[0.43243243 0.]
[0.7027027 0.48059634]
[0. 0.01650165]
[0.64864865 0.21748037]
[0.10810811 0.21748037]
[0.40540541 0.2208945]
[0.51351351 0.44508934]
[0.10810811 0.48059634]
[0.45945946 0.48059634]
[0.40540541 0.22260157]
[1. 0.23842039]
[0.13513514 0.42505975]
[0.40540541 0.54011608]
[0.40540541 0.68225788]
[0.13513514 0.89382042]
[0.43243243 0.82576534]
[0.72972973 0.93262775]
[0.13513514 0.48059634]
[0.40540541 0.10754524]
[0.72972973 0.48059634]
[0.13513514 0.20996927]
[0.72972973 0.33515421]
[0.43243243 0.44895869]
[0.13513514 0.80949129]
[0.40540541 0.90303858]
[0.43243243 0.48059634]
[0.43243243 0.48059634]
[0.43243243 0.13781723]
[0.40540541 0.24411062]

```
[0.13513514 0.36815751]
[0.40540541 0.49459429]
[0.40540541 0.50825083]
[0.13513514 0.87606692]
[0.43243243 0.87606692]
[0.72972973 1.         ]
[0.13513514 0.48059634]
[0.43243243 0.         ]
[0.72972973 0.48059634]
[0.40540541 0.01650165]
[0.72972973 0.24411062]
[0.40540541 0.2691476  ]
[0.13513514 0.87606692]
[0.43243243 0.51951747]
[0.08108108 0.44611358]
[0.35135135 0.45624218]
[0.43243243 0.22260157]
[0.89189189 0.23842039]
[0.10810811 0.42505975]
[0.40540541 0.54011608]
[0.40540541 0.68225788]
[0.13513514 0.89382042]
[0.37837838 0.82576534]
[0.62162162 0.93262775]
[0.40540541 0.48059634]
[0.37837838 0.10754524]
[0.72972973 0.22123592]
[0.13513514 0.20996927]
[0.72972973 0.33515421]
[0.43243243 0.44895869]
[0.13513514 0.80949129]
[0.43243243 0.90303858]
[0.13513514 0.87606692]
[0.72972973 0.33515421]
[0.43243243 0.44895869]
[0.13513514 0.80949129]]
```

```
In [28]: mm_scaled_data_set = pd.DataFrame(data_scaled_mm,columns=data.columns)
mm_scaled_data_set
```

Out[28]:

	Age	Salary
0	0.081081	0.480596
1	0.351351	0.480596
2	0.486486	0.137817
3	0.621622	0.217480
4	0.162162	0.331285
...	...	...
143	0.432432	0.903039
144	0.135135	0.876067
145	0.729730	0.335154
146	0.432432	0.448959
147	0.135135	0.809491

148 rows × 2 columns

In [32]: `mm_scaled_data_set.to_csv("mm_scaled_data_set.csv")`